From Linear Models to GAM

Understanding Non-Linearity in Global Health and One Health Systems

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1. Introduction: The Illusion of Linearity

We often start with a simple assumption:

> Increasing environmental exposure leads to proportional increase in health risk.

This naturally leads to linear regression.

However, environmental and epidemiological systems are rarely linear.

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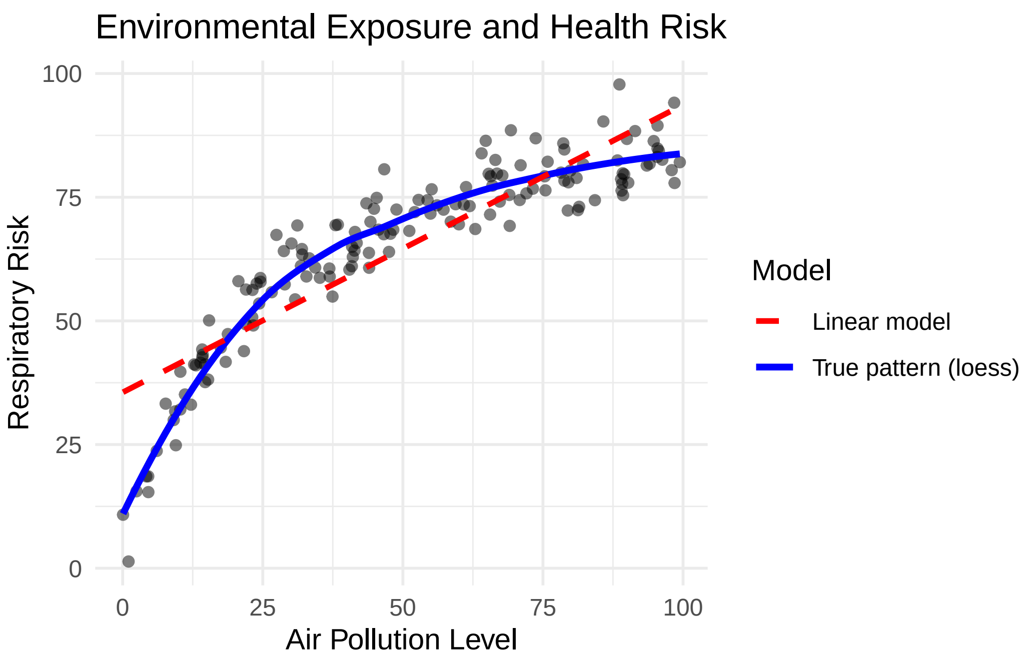
2. Simulating an Environmental Health Reality

```r
set.seed(123)

n <- 150

air_pollution_level <- runif(n, 0, 100)

max_population_risk <- 100
exposure_half_risk <- 20
```

```r
respiratory_risk <-
  (max_population_risk * air_pollution_level) /
  (exposure_half_risk + air_pollution_level) +
  rnorm(n, 0, 5)

df <- data.frame(
  air_pollution_level,
  respiratory_risk
)

ggplot(df, aes(air_pollution_level, respiratory_risk)) +
  geom_point(alpha = 0.5) +

  geom_smooth(
    aes(color = "True pattern (loess)"),
    method = "loess",
    se = FALSE,
    size = 1.2
  ) +

  geom_smooth(
    aes(color = "Linear model"),
    method = "lm",
    formula = y ~ x,
    linetype = "dashed",
    se = FALSE
  ) +

  scale_color_manual(
    name = "Model",
    values = c(
      "True pattern (loess)" = "blue",
      "Linear model" = "red"
    )
  ) +

  labs(
    title = "Environmental Exposure and Health Risk",
    x = "Air Pollution Level",
    y = "Respiratory Risk"
  ) +
```

```
  theme_minimal()
```



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3. The Wrong Model: Linear Regression

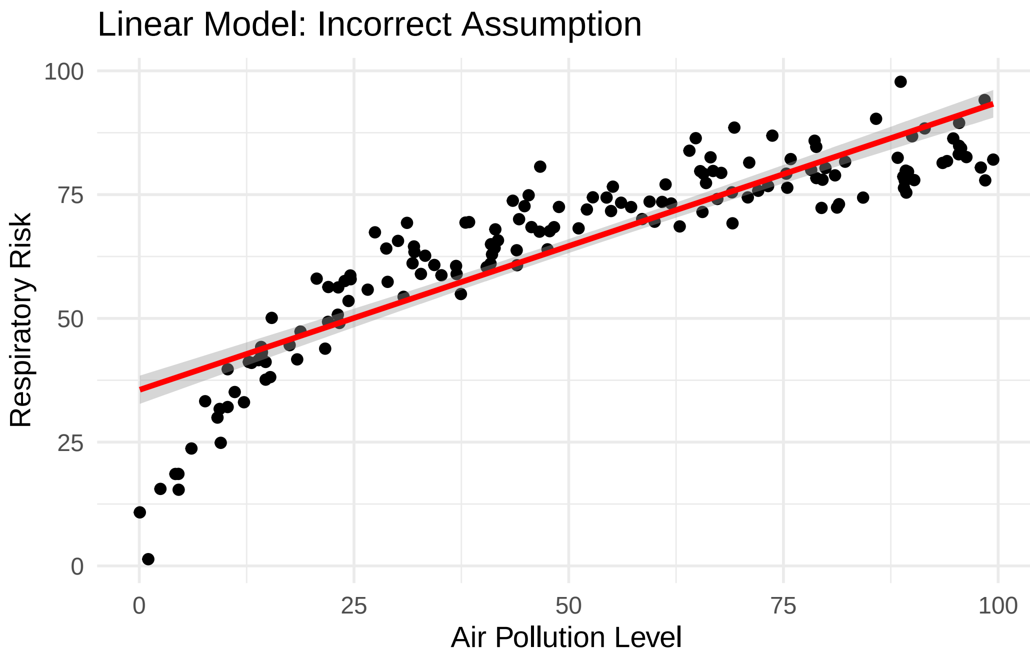
The linear model assumes:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

This implies a constant increase in health risk for each additional unit of environmental exposure.

```
model_lm_simple <- lm(
  respiratory_risk ~ air_pollution_level,
  data = df
)

ggplot(df, aes(air_pollution_level, respiratory_risk)) +
  geom_point() +
```

```r
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Linear Model: Incorrect Assumption",
    x = "Air Pollution Level",
    y = "Respiratory Risk"
  ) +
  theme_minimal()
```




```r
summary(model_lm_simple)
```

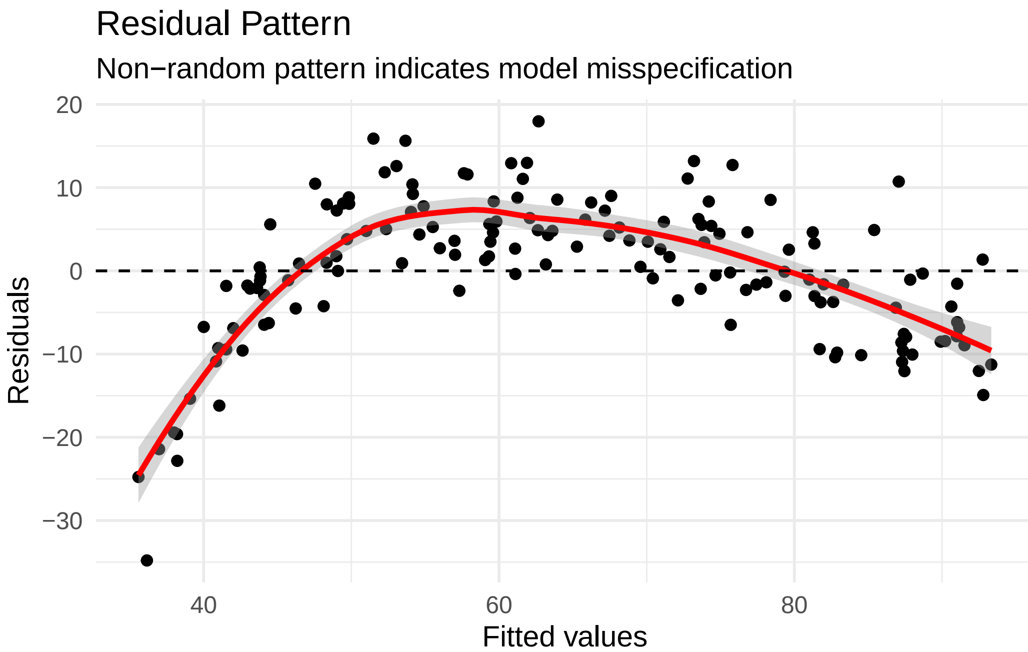
```
Call:
lm(formula = respiratory_risk ~ air_pollution_level, data = df)

Residuals:
    Min      1Q  Median      3Q     Max 
-34.799  -4.496   0.899   5.621  17.970 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         35.54817    1.44292   24.64   <2e-16 ***
air_pollution_level  0.58120    0.02489   23.35   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.724 on 148 degrees of freedom
Multiple R-squared:  0.7865,    Adjusted R-squared:  0.7851
F-statistic: 545.3 on 1 and 148 DF,  p-value: < 2.2e-16
```

---

4. The Diagnostic: Residuals

```r
ggplot(model_lm_simple, aes(.fitted, .resid)) +
  geom_point() +
  geom_smooth(method = "loess", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Residual Pattern",
    subtitle = "Non-random pattern indicates model misspecification",
    x = "Fitted values",
    y = "Residuals"
  ) +
  theme_minimal()
```



Residuals are not just diagnostics — they often reveal when the model is misspecified.

A correct model should produce residuals randomly scattered around zero.

---

5. Polynomial Regression

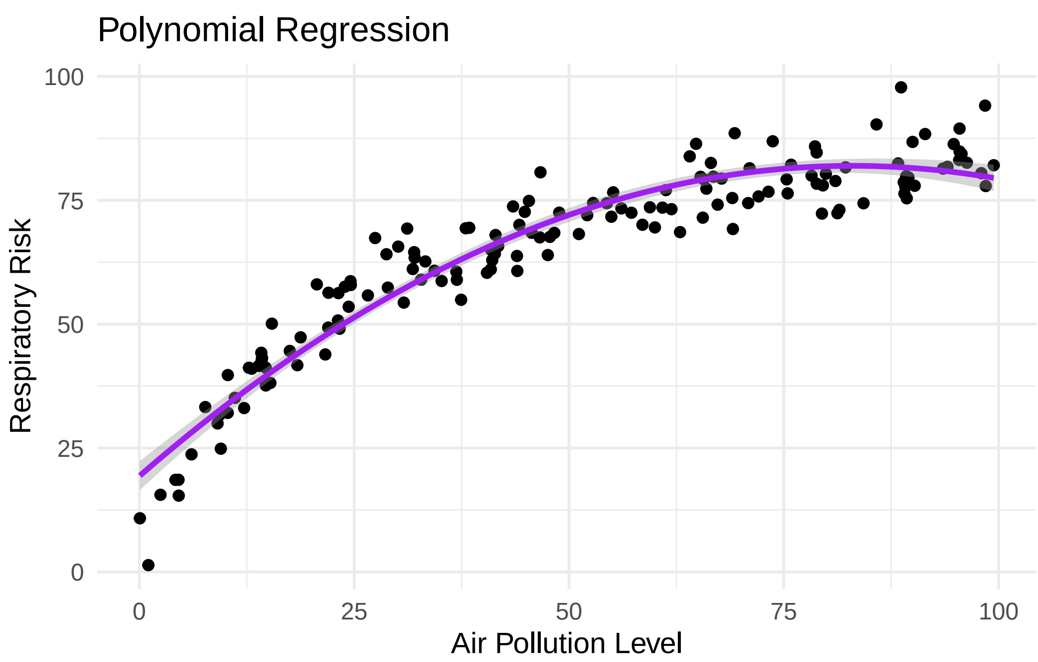
To capture curvature, we extend the linear model:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

This allows the model to bend, but it still does not represent the true environmental mechanism.

```r
model_poly <- lm(
  respiratory_risk ~ air_pollution_level + I(air_pollution_level^2),
  data = df
)

ggplot(df, aes(air_pollution_level, respiratory_risk)) +
  geom_point() +
```

```
  geom_smooth(
    method = "lm",
    formula = y ~ x + I(x^2),
    color = "purple"
  ) +
  labs(
    title = "Polynomial Regression",
    x = "Air Pollution Level",
    y = "Respiratory Risk"
  ) +
  theme_minimal()
```




```
summary(model_poly)
```

```

Call:
lm(formula = respiratory_risk ~ air_pollution_level + I(air_pollution_level^2),
    data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-19.5376  -3.6698  -0.3763   3.6491  16.1271
```

```
Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)              19.3266118  1.4893567   12.98   <2e-16 ***
air_pollution_level       1.5084388  0.0678113   22.25   <2e-16 ***
I(air_pollution_level^2) -0.0090828  0.0006448  -14.09   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.711 on 147 degrees of freedom
Multiple R-squared:  0.9092,    Adjusted R-squared:  0.9079
F-statistic: 735.6 on 2 and 147 DF,  p-value: < 2.2e-16
```

Polynomial regression improves the fit by introducing curvature, but it remains a statistical approximation.

Although the model captures non-linearity, it introduces an artificial decline in risk at high exposure levels, which is not epidemiologically plausible.

> A good fit does not imply a correct model.

Interpretation

The fitted model can be written as:

$$Risk = 19.33 + 1.51(Exposure) - 0.009(Exposure^2)$$

- The positive linear term indicates an initial increase in health risk.
- The negative quadratic term introduces curvature and diminishing returns.

The model predicts a maximum risk at:

$$Exposure_{max} = \frac{-\beta_1}{2\beta_2}$$

However, the decline beyond this point is artificial and does not reflect the true exposure–risk mechanism.

---

6. Model Comparison

6.1 ANOVA Comparison

```r
model_base <- lm(
  respiratory_risk ~ air_pollution_level,
  data = df
)

model_extended <- lm(
  respiratory_risk ~ air_pollution_level + I(air_pollution_level^2),
  data = df
)

anova(model_base, model_extended)
```

```
Analysis of Variance Table

Model 1: respiratory_risk ~ air_pollution_level
Model 2: respiratory_risk ~ air_pollution_level + I(air_pollution_level^2)
  Res.Df     RSS Df Sum of Sq      F    Pr(>F)
1    148 11264.3
2    147  4793.6  1    6470.6 198.43 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

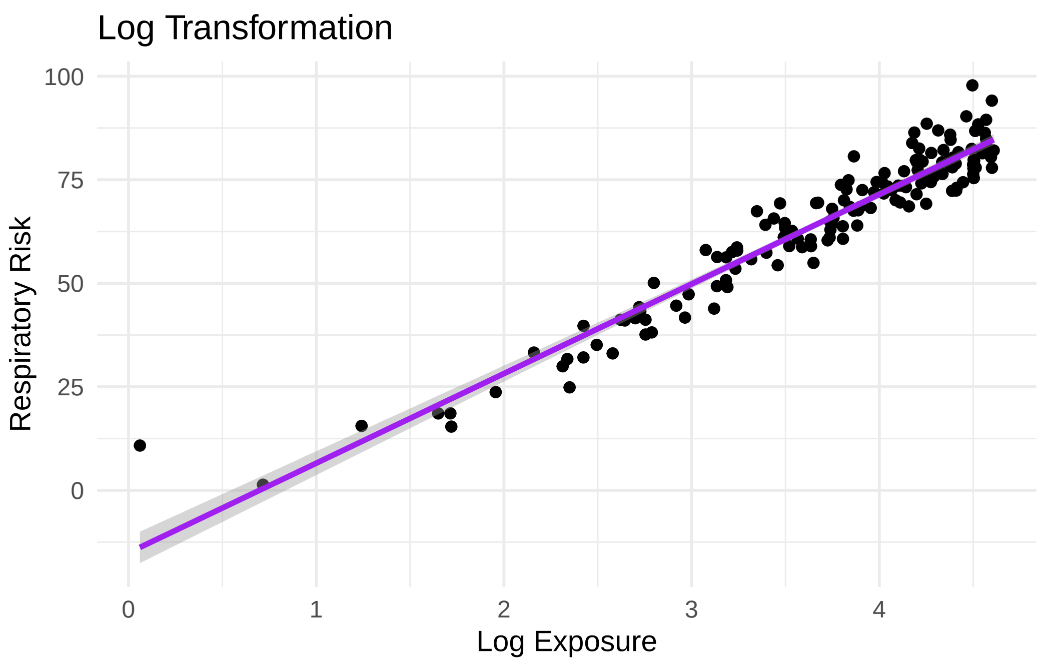
- p-value $< 0.05 \rightarrow$ the additional term improves the model
- p-value $> 0.05 \rightarrow$ the simpler model is sufficient

---

7. The Temptation: Transformations

```r
df_log <- df %>%
  mutate(log_exposure = log(air_pollution_level + 1))
```

```r
model_log <- lm(
  respiratory_risk ~ log_exposure,
  data = df_log
)

ggplot(df_log, aes(log_exposure, respiratory_risk)) +
  geom_point() +
  geom_smooth(method = "lm", color = "purple") +
  labs(
    title = "Log Transformation",
    x = "Log Exposure",
    y = "Respiratory Risk"
  ) +
  theme_minimal()
```



Why transformations can fail

- They distort interpretation
- They may introduce heteroscedasticity
- They force the data into a linear structure

---

8. Non-linear Least Squares (NLS)

Instead of transforming the data, we directly model the environmental mechanism:

$$Risk = \frac{R_{max} \cdot Exposure}{E_{50} + Exposure} + \varepsilon$$

Where:

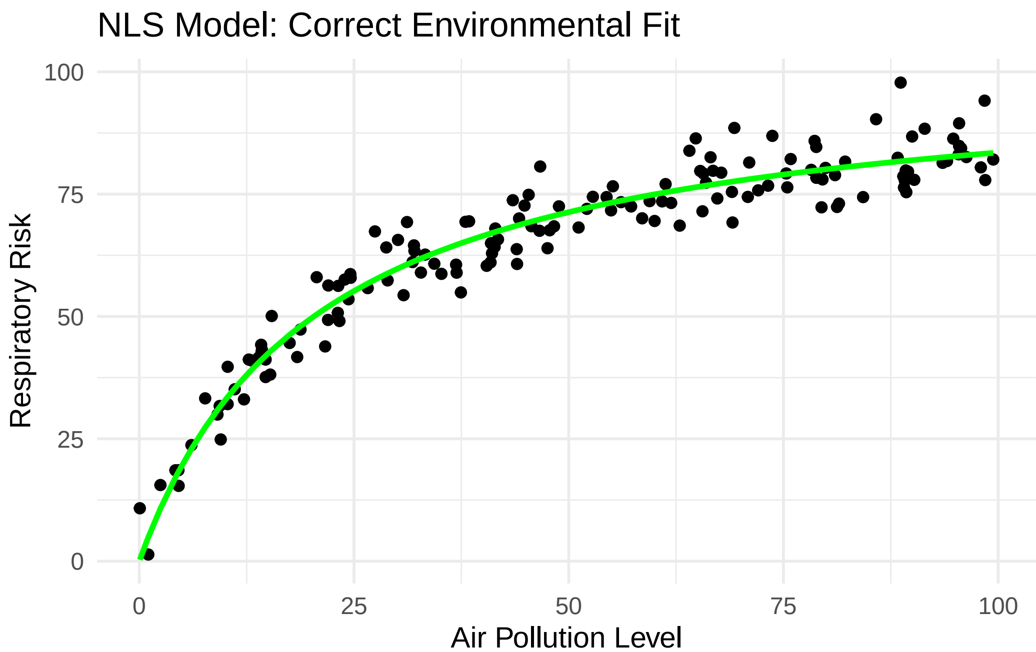
- (R_{max}): maximum achievable population risk
- (E_{50}): exposure level associated with 50% of the maximum risk

```r
model_nls <- nls(
  respiratory_risk ~
    (Rmax * air_pollution_level) /
    (E50 + air_pollution_level),
  data = df,
  start = list(Rmax = 80, E50 = 10)
)

summary(model_nls)
```

```
Formula: respiratory_risk ~ (Rmax * air_pollution_level)/(E50 + air_pollution_level)

Parameters:
     Estimate Std. Error t value Pr(>|t|)
Rmax  100.702      1.575   63.94   <2e-16 ***
E50    20.594      1.075   19.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.777 on 148 degrees of freedom

Number of iterations to convergence: 4
Achieved convergence tolerance: 3.112e-06
```

```r
df$pred_nls <- predict(model_nls)

ggplot(df, aes(air_pollution_level, respiratory_risk)) +
  geom_point() +
  geom_line(aes(y = pred_nls), color = "green", size = 1) +
```

```r
  labs(
    title = "NLS Model: Correct Environmental Fit",
    x = "Air Pollution Level",
    y = "Respiratory Risk"
  ) +
  theme_minimal()
```



---

9. Other Non-linear Forms

Exponential Model

$$Y = ae^{bX}$$

Used for epidemic growth or infectious disease propagation.

```r
model_exp <- nls(
  respiratory_risk ~ a * exp(b * air_pollution_level),
  data = df,
  start = list(a = 1, b = 0.01)
```

```r
)

summary(model_exp)
```

```
Formula: respiratory_risk ~ a * exp(b * air_pollution_level)

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 4.202e+01  1.312e+00   32.04   <2e-16 ***
b 8.126e-03  4.533e-04   17.93   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.05 on 148 degrees of freedom

Number of iterations to convergence: 9
Achieved convergence tolerance: 3.596e-06
```

Power Model

$$Y = aX^b$$

Common in environmental scaling relationships.

```r
model_power <- nls(
  respiratory_risk ~ a * air_pollution_level^b,
  data = df,
  start = list(a = 1, b = 1)
)

summary(model_power)
```

```
Formula: respiratory_risk ~ a * air_pollution_level^b

Parameters:
  Estimate Std. Error t value Pr(>|t|)
```

```
a 16.11420    0.79864   20.18   <2e-16 ***
b  0.36944    0.01218   30.33   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.671 on 148 degrees of freedom

Number of iterations to convergence: 13
Achieved convergence tolerance: 5.405e-06
```

Logarithmic Model

$$Y = a \log(X)$$

Represents rapid initial increase followed by slowing growth.

```r
model_log2 <- nls(
  respiratory_risk ~ a * log(air_pollution_level + 1),
  data = df,
  start = list(a = 10)
)

summary(model_log2)
```

```
Formula: respiratory_risk ~ a * log(air_pollution_level + 1)

Parameters:
  Estimate Std. Error t value Pr(>|t|)
a  17.7529     0.1338   132.7   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.206 on 149 degrees of freedom

Number of iterations to convergence: 1
Achieved convergence tolerance: 4.584e-09
```

9.1 Self-Starting Non-linear Models

In practice, choosing starting values manually can be difficult.

To simplify this process, R provides several self-starting non-linear models (`SS*`) that automatically estimate reasonable initial values from the data.

This reduces convergence problems and simplifies model fitting.

```r
model_auto <- nls(
  respiratory_risk ~
    SSmicmen(air_pollution_level, Rmax, E50),
  data = df
)

summary(model_auto)
```

```
Formula: respiratory_risk ~ SSmicmen(air_pollution_level, Rmax, E50)

Parameters:
     Estimate Std. Error t value Pr(>|t|)
Rmax  100.702      1.575   63.94   <2e-16 ***
E50    20.594      1.075   19.15   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.777 on 148 degrees of freedom

Number of iterations to convergence: 0
Achieved convergence tolerance: 8.06e-08
```

The function `SSmicmen()` implements the Michaelis–Menten model:

$$Y = \frac{R_{max} \cdot X}{E_{50} + X}$$

but automatically estimates starting values before optimization begins.

This is particularly useful in epidemiology and environmental health studies where plausible parameter values may not be known in advance.

> Self-starting models improve convenience, but they do not eliminate all convergence problems.

If the data are highly noisy, sparse, or poorly scaled, optimization may still fail.

---

10. Model Comparison

```r
data.frame(
  Model = c(
    "Linear",
    "Polynomial",
    "Log",
    "NLS",
    "Exponential",
    "auto-NLS"
  ),
  AIC = c(
    AIC(model_lm_simple),
    AIC(model_poly),
    AIC(model_log),
    AIC(model_nls),
    AIC(model_exp),
    AIC(model_auto)
  )
)
```

```
        Model       AIC
1      Linear 1079.4949
2  Polynomial  953.3432
3         Log  927.3915
4         NLS  898.8311
5 Exponential 1121.8168
6    auto-NLS  898.8311
```

Interpretation

- Lower AIC indicates a better balance between fit and complexity.
- Large AIC differences suggest structural misspecification.

---

11. GAM: Flexible Non-linear Modeling

What if we do not know the true functional form?

Instead of imposing a structure, GAM learns it directly from the data.

11.1 Complex Environmental Signal

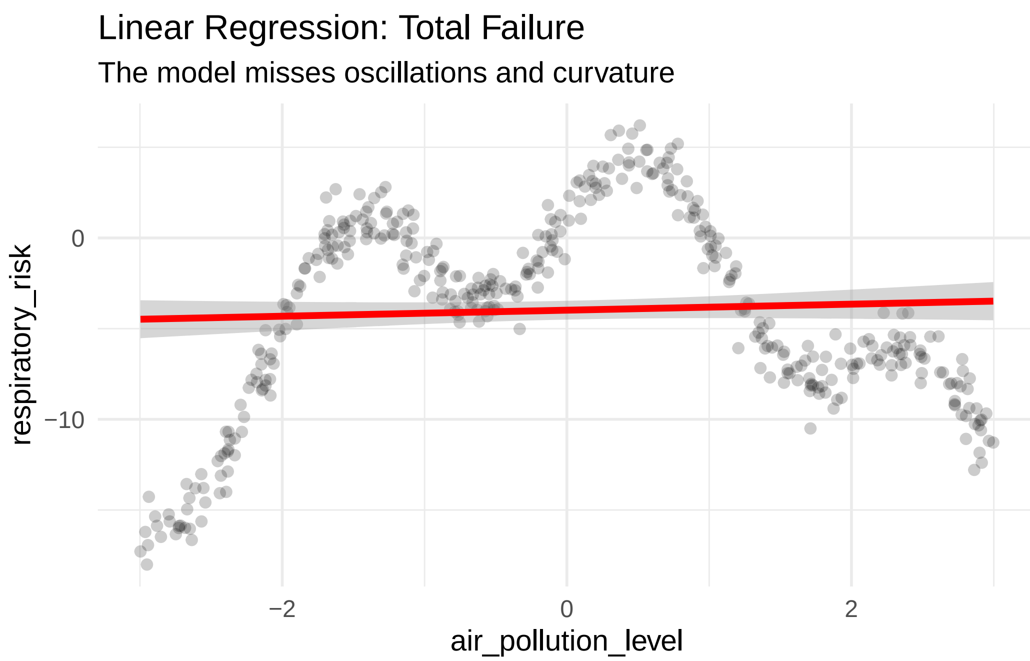
```
set.seed(123)

n <- 400

air_pollution_level <- runif(n, -3, 3)

noise <- rnorm(n, 0, 1)

respiratory_risk <-
  4 * sin(3 * air_pollution_level) +
  cos(0.8 * air_pollution_level) -
  1.5 * (air_pollution_level^2) +
  noise

df_gam <- data.frame(
  air_pollution_level,
  respiratory_risk
)
```

This combines:

- seasonal environmental fluctuations
- epidemic waves
- long-term population trends

11.2 Linear Regression Failure

```r
model_lm_complex <- lm(
  respiratory_risk ~ air_pollution_level,
  data = df_gam
)

ggplot(df_gam, aes(air_pollution_level, respiratory_risk)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "red", size = 1.2) +
  labs(
    title = "Linear Regression: Total Failure",
    subtitle = "The model misses oscillations and curvature"
  ) +
  theme_minimal()
```



11.3 The GAM Idea

$$Y = \beta_0 + s(X) + \varepsilon$$

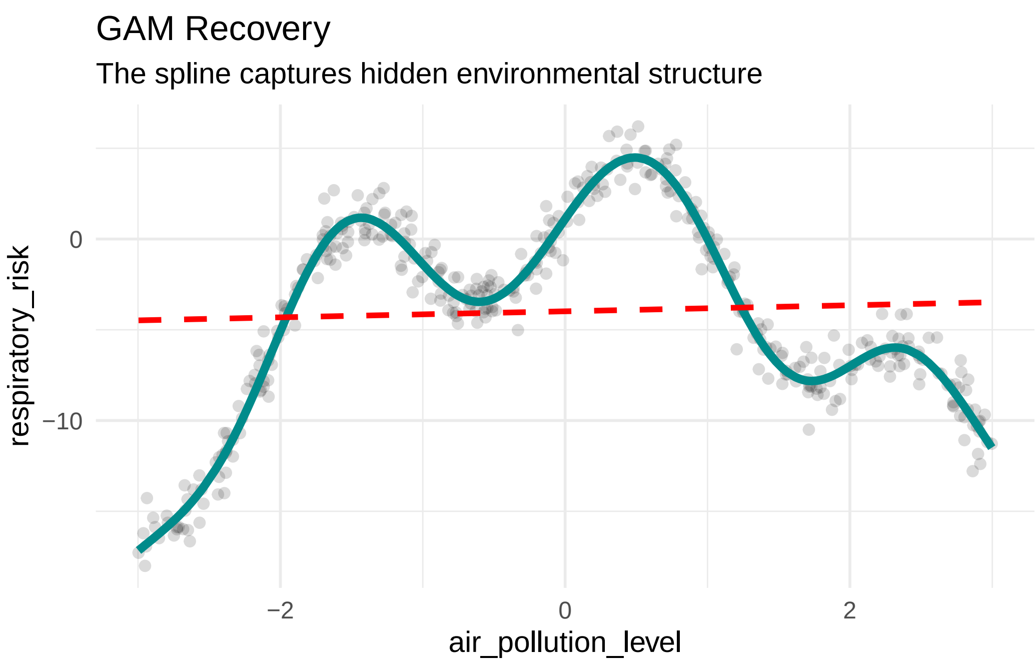
Where:

- (s(X)) is a smooth function estimated from the data.

11.4 GAM Fit

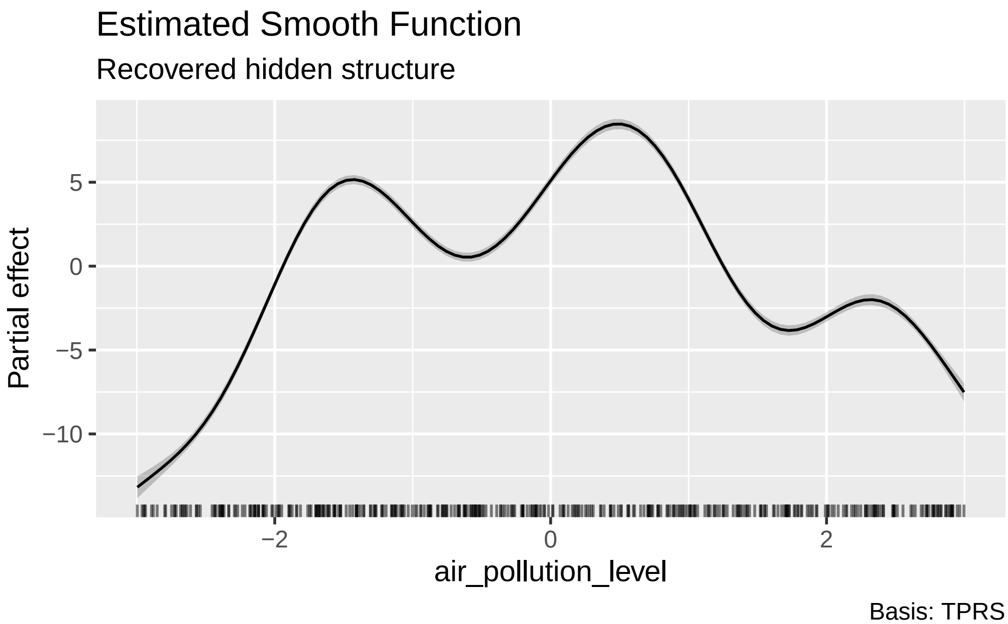
```r
model_gam_complex <- gam(
  respiratory_risk ~ s(air_pollution_level),
  data = df_gam,
  method = "REML"
)

df_gam$pred_gam <- predict(model_gam_complex)
```

```r
ggplot(df_gam, aes(air_pollution_level, respiratory_risk)) +
  geom_point(alpha = 0.15) +
  geom_line(aes(y = pred_gam), color = "cyan4", size = 1.5) +
  geom_smooth(
    method = "lm",
    color = "red",
    linetype = "dashed",
    se = FALSE
  ) +
  labs(
    title = "GAM Recovery",
    subtitle = "The spline captures hidden environmental structure"
  ) +
  theme_minimal()
```



11.5 Smooth Function Visualization

```
draw(model_gam_complex) +
  labs(
    title = "Estimated Smooth Function",
    subtitle = "Recovered hidden structure"
  )
```



11.6 Model Comparison

```r
results <- data.frame(
  Model = c("Linear Model", "GAM"),
  Adjusted_R2 = c(
    summary(model_lm_complex)$adj.r.squared,
    summary(model_gam_complex)$r.sq
  ),
  AIC = c(
    AIC(model_lm_complex),
    AIC(model_gam_complex)
  )
)

kable(results, digits = 3)
```

Model	Adjusted_R2	AIC
Linear Model	0.000	2468.798
GAM	0.965	1136.281

Interpretation

- Linear model → global trend only
- GAM → local + global structure

  GAM adapts to the data instead of forcing a predefined shape.

---

12. GAM with Multiple Predictors

A hybrid GAM combines smooth and linear terms:

$$Y = \beta_0 + s(X_1) + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

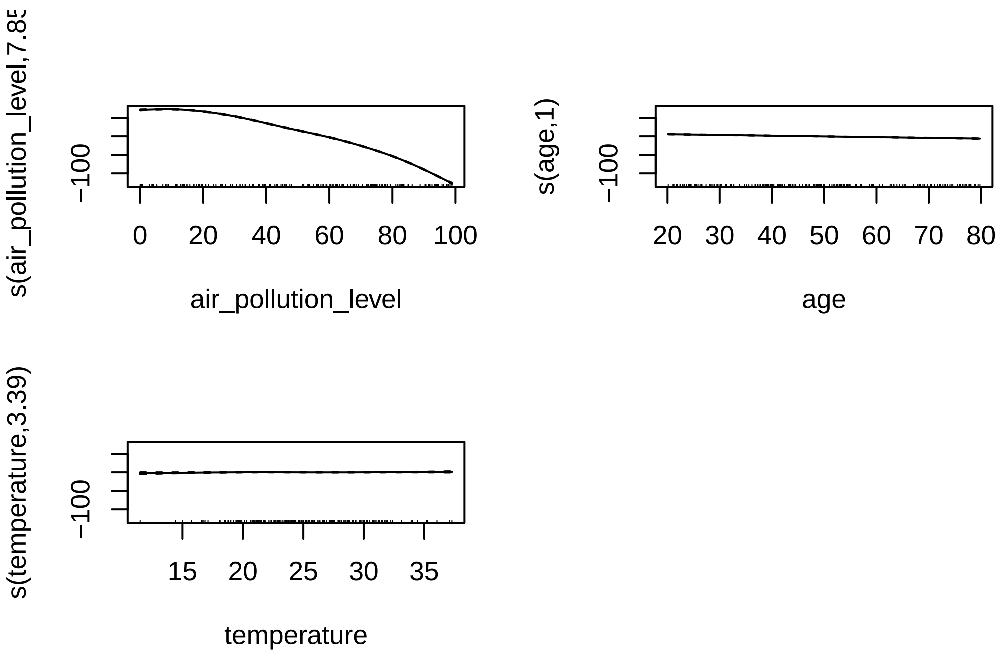
12.1 Simulation

```
set.seed(42)

n <- 200

air_pollution_level <- runif(n, 0, 100)

age <- runif(n, 20, 80)

temperature <- rnorm(n, 25, 5)

respiratory_risk <-
  4 * sin(air_pollution_level / 10) +
```

```
  cos(air_pollution_level / 20) -
  0.02 * air_pollution_level^2 +
  (-0.2 * age) +
  rnorm(n, 0, 2)

df2 <- data.frame(
  air_pollution_level,
  age,
  temperature,
  respiratory_risk
)
```

12.2 Wrong Approach: Smoothing Everything

```
gam_all <- gam(
  respiratory_risk ~
    s(air_pollution_level) +
    s(age) +
    s(temperature),
  data = df2
)

plot(gam_all, pages = 1)
```

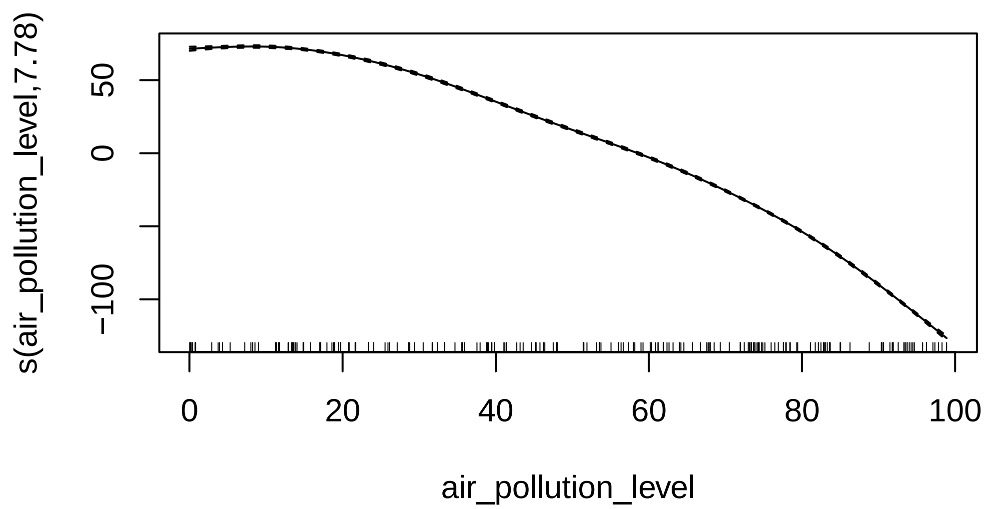


Problem

- unnecessary complexity
- reduced interpretability
- not all variables require smoothing

12.3 Hybrid GAM

```
gam_hybrid <- gam(
  respiratory_risk ~
    s(air_pollution_level) +
    age +
    temperature,
  data = df2
)

plot(gam_hybrid, shade = TRUE)
```



12.4 Interpretation

```r
summary(gam_hybrid)
```

```
Family: gaussian
Link function: identity

Formula:
respiratory_risk ~ s(air_pollution_level) + age + temperature

Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -72.097913   0.965595 -74.667   <2e-16 ***
age          -0.195943   0.008637 -22.686   <2e-16 ***
temperature   0.039203   0.032906   1.191    0.235
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Approximate significance of smooth terms:
                        edf Ref.df     F p-value
s(air_pollution_level) 7.781  8.614 18969  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.999   Deviance explained = 99.9%
GCV = 4.6565  Scale est. = 4.4055    n = 200
```

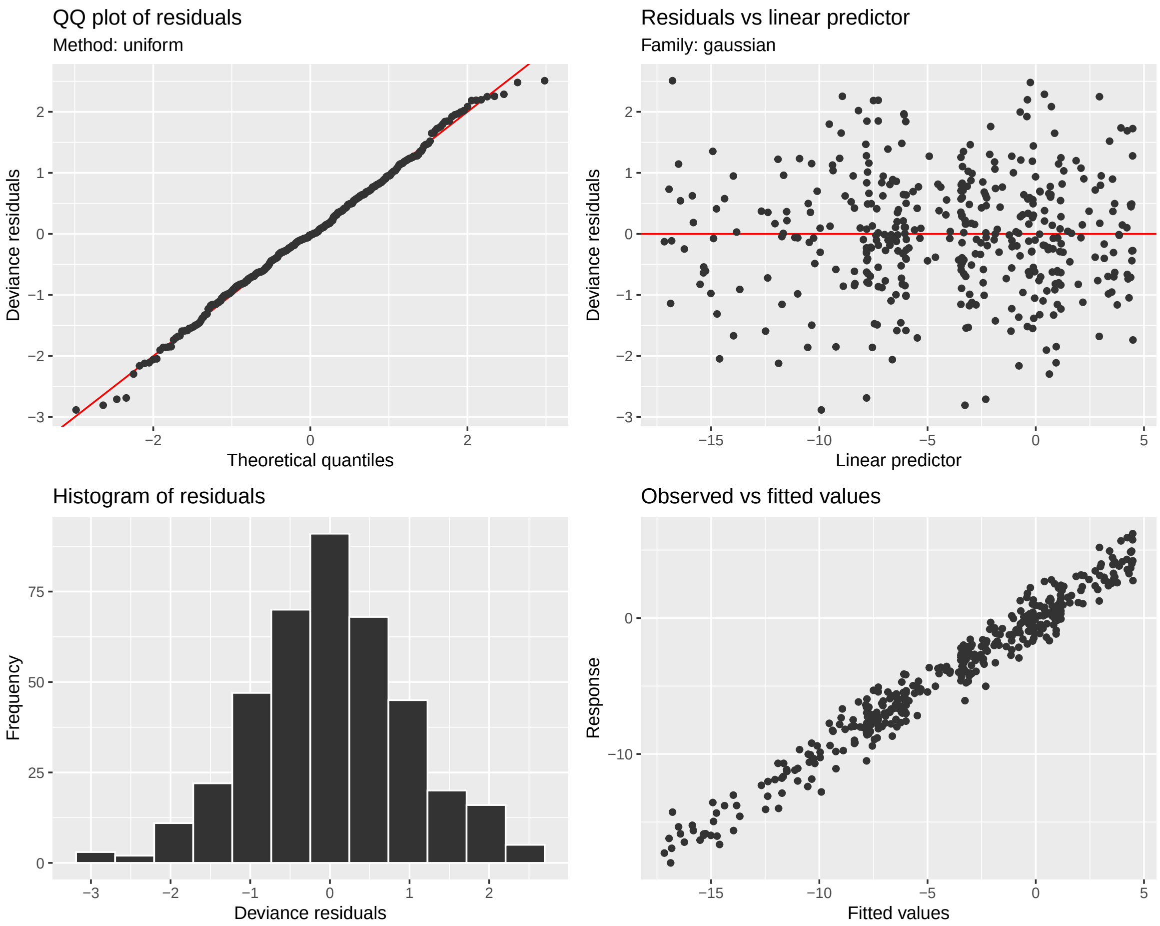
Interpretation of effective degrees of freedom (edf):

- edf 1 $\rightarrow$ approximately linear
- edf $> 1 \rightarrow$ non-linear relationship

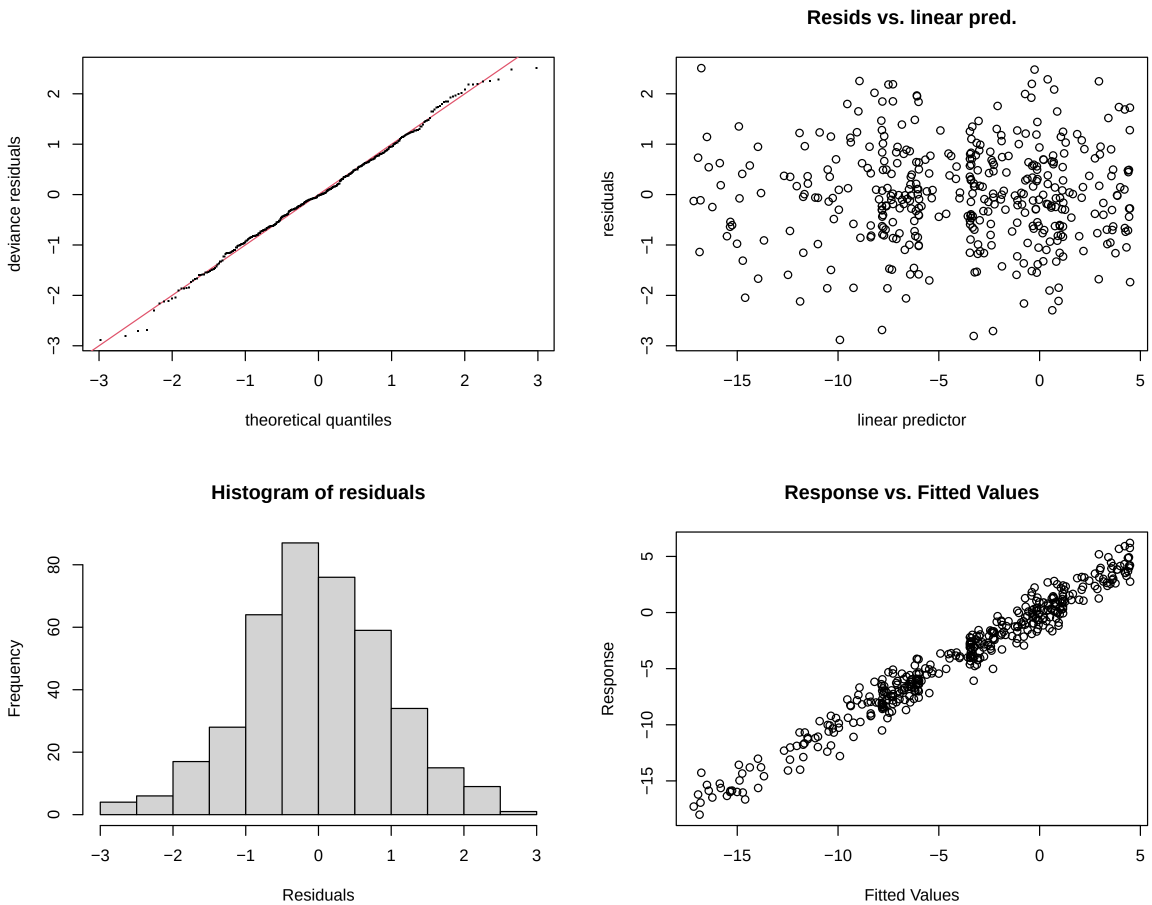
  Because the data were simulated from a highly structured non-linear process, the GAM is able to recover almost the entire signal. In real epidemiological datasets, performance is usually lower due to measurement error, unmeasured confounding, and population heterogeneity.

12.5 Diagnostics

```r
appraise(model_gam_complex)
```

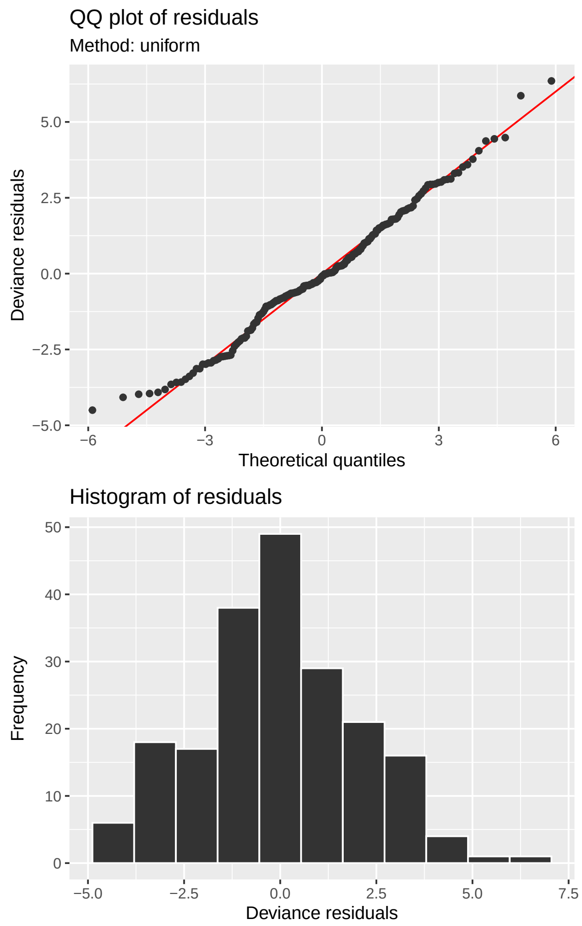



```r
gam.check(model_gam_complex)
```

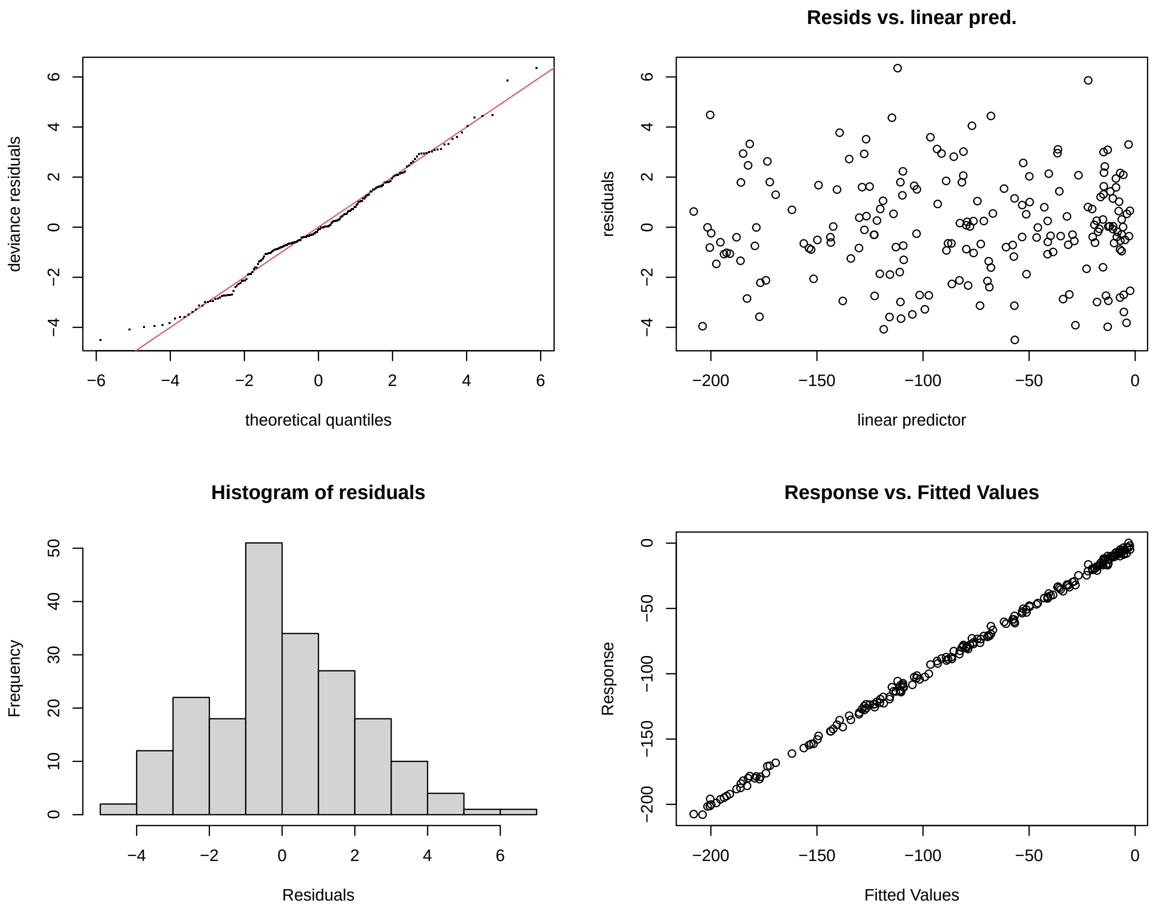



```
Method: REML   Optimizer: outer newton
full convergence after 8 iterations.
Gradient range [-5.332214e-05,5.260707e-05]
(score 593.2697 & scale 0.9735279).
Hessian positive definite, eigenvalue range [3.884655,199.0815].
Model rank =  10 / 10

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                        k'  edf k-index p-value
s(air_pollution_level) 9.00 8.97    1.12    0.98
```

```r
appraise(gam_hybrid)
```




```r
gam.check(gam_hybrid)
```




```
Method: GCV   Optimizer: magic
Smoothing parameter selection converged after 10 iterations.
The RMS GCV score gradient at convergence was 9.068241e-06 .
The Hessian was positive definite.
Model rank =  12 / 12

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                        k'  edf k-index p-value
s(air_pollution_level) 9.00 7.78    1.04    0.69
```

These diagnostics evaluate:

- residual structure
- adequacy of smoothing
- model assumptions

12.6 Tensor Product Smooths: Joint Non-linear Effects

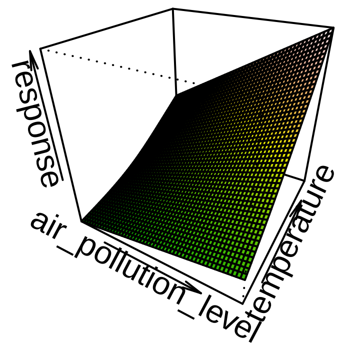
```
set.seed(123)

n <- 300

air_pollution_level <- runif(n, 0, 100)

temperature <- runif(n, 10, 40)

respiratory_risk <-
  0.03 * air_pollution_level^1.2 +
  0.04 * temperature^2 +
  0.02 * air_pollution_level * temperature +
  rnorm(n, 0, 3)

df_tensor <- data.frame(
  air_pollution_level,
  temperature,
  respiratory_risk
)
```

Tensoriel GAM

```
model_tensor <- gam(
  respiratory_risk ~
    te(air_pollution_level, temperature),
  data = df_tensor,
  method = "REML"
)

summary(model_tensor)
```

```
Family: gaussian
Link function: identity

Formula:
respiratory_risk ~ te(air_pollution_level, temperature)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  55.9842     0.1717   326.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                      edf Ref.df    F p-value
te(air_pollution_level,temperature) 8.53  9.871 3095  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =   0.99   Deviance explained = 99.1%
-REML = 755.36  Scale est. = 8.8415    n = 300
```

```r
vis.gam(
  model_tensor,
  type = "response",
  plot.type = "persp",
  phi = 30,
  theta = 30,
  n.grid = 50,
  color = "terrain"
)
```



The surface plot reveals that respiratory risk is not determined by pollution or temperature independently.

Instead, the risk emerges from their interaction:

high pollution combined with high temperature produces the highest risk, while moderate exposure levels may have limited impact.

This illustrates a core principle of modern epidemiology:

health outcomes often emerge from interacting systems rather than isolated variables.

12.7 Common Diagnostic Problems in GAMs

Model diagnostics are not merely technical checks.

They help identify: - structural misspecification, - distributional problems, - overfitting, - insufficient smoothing, - and violations of statistical assumptions.

Diagnostic Issue	Visual Symptom	Statistical Interpretation	Corrective Action
Non-linearity	Patterns or "waves" in residuals vs fitted plot	The model misses structural complexity (underfitting)	Increase basis dimension (`k`) or add non-linear terms
Heteroscedasticity	"Fan" or "funnel" shape in residuals	Non-constant variance	Use another distribution family (`Gamma`, `nb`) or transform the response
Non-normality	QQ-plot strongly deviates from the line	Residual distribution is skewed or heavy-tailed	Investigate outliers or consider robust distributions
Autocorrelation	Residual clustering across time or space	Violation of independence assumption	Use GAMM with correlation structures or spatial effects
Under-smoothing	Low p-value in `gam.check()` (`k-index < 1`)	Basis dimension too small	Increase `k`
Overfitting	Very high edf with poor generalization	Model captures noise instead of signal	Increase smoothing penalty (`gamma`) or reduce `k`

12.8 Understanding Basis Functions in GAMs

In GAMs, smooth functions are constructed using basis functions (`bs`).

The default basis in `mgcv` is:

```
bs = "tp"
```

which stands for **thin plate regression splines**.

Thin plate splines are widely used because they:

- do not require manual knot placement,
- are generally stable and flexible,
- work well as general-purpose smoothers,
- automatically balance flexibility and smoothness.

Common Alternative Bases

Basis	Meaning	Main Idea
`"tp"`	Thin plate spline	Default general-purpose smoother
`"cr"`	Cubic regression spline	Faster computation with fixed knots
`"cs"`	Cubic spline with shrinkage	Allows unnecessary smooth terms to shrink toward zero
`"ps"`	Penalized spline	Flexible smoothing with explicit penalty control
`"ds"`	Duchon spline	Generalization of thin plate splines for complex multidimensional smoothing

---

Interaction Smooths in GAMs

Instead of a simple smooth term:

```
s(x)
```

GAMs also allow smooth interactions between variables.

Common Interaction Terms

Function	Interpretation
`te(x, z)`	Main smooth effects + interaction between `x` and `z`
`ti(x, z)`	Interaction smooth only
`t2(x, z)`	Advanced tensor-product smooth with more flexible penalty structure

Conceptual Difference

- `te()` models both the individual effects and their interaction.
- `ti()` isolates only the interaction component.
- `t2()` is mainly used in advanced GAM formulations requiring complex penalty decomposition.

---

13. Final Takeaway

- Linear models are often insufficient in environmental and epidemiological systems.
- Transformations may distort interpretation.
- NLS models respect mechanistic relationships.
- GAM provides flexibility when the functional form is unknown.
- Hybrid GAMs balance interpretability and flexibility.

Final Message

> The goal is not to fit the simplest model, but the model that best represents the underlying environmental and epidemiological mechanism.
>
> In Global Health and One Health research, non-linearity is the rule rather than the exception.